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Dear Editors,

On behalf of all co-authors (Rodrigo Almeida de Oliveira, Juliano de Paulo Ribeiro, and Prof. Edson Emilio Scalabrin), I am pleased to submit our manuscript titled “**Process Mining and Stochastic Modeling for Software Process Forecasting: A Systematic Literature Review with an LLM-Assisted Protocol**” for consideration in *Information and Software Technology*.

Novelty and contribution. This manuscript reports a systematic literature review (SLR) at the intersection of process mining (PM), stochastic modeling, and machine learning applied to software process forecasting. The contribution is threefold:

1. **Methodological:** we operationalise an LLM-assisted screening protocol with cross-model verification (Claude Haiku 4.5 as primary screener, Claude Sonnet 4.6 as verifier), achieving substantial Cohen’s κ (binary $\kappa = 0.695$ for title/abstract, 0.694 for full-text) on a stratified random 20% sample. The protocol extends the recent work of Khraisha et al. (2024) and Syriani et al. (2023) to the software-process-mining domain at a scale of 6,147 deduplicated papers across two corpus tiers.
2. **Empirical:** we confirm 404 primary studies (169 working-set + 212 + 23 auxiliary) covering 1995–2026, scored against an 8-criterion quality rubric (Dyba and Dingsoyr, 2008). The combined mean QA score is 5.00 ± 1.42 (315 papers, 82.7%, retained for synthesis).
3. **Synthetic:** we report five empirically grounded findings (F1–F5), propose the SPMF taxonomy (Software Process Mining and Forecasting) along three orthogonal dimensions, and articulate a four-direction research agenda (RA1–RA4). The IC-combination analysis shows that only 1 of 404 papers (0.3%) jointly satisfies process mining, stochastic modeling, and forecasting — evidence that the gap addressed by our PATHCAST proposal is structural rather than an artefact of the search queries.

Fit with IST scope. The manuscript squarely fits IST’s scope on empirical software engineering methods, mining software repositories, and process modeling. It complements the journal’s growing body of evidence-based SLRs while contributing to the methodological literature on LLM-assisted reviews.

Reproducibility. A complete replication package is deposited at Zenodo (DOI 10.5281/zenodo.20130276), including all search strings, screening prompts, JSON schemas, decision CSVs at every pipeline stage, the cross-model κ scripts, and the structured extraction tables.

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Prior submission and conflicts. The manuscript is original work, has not been published elsewhere, and is not under consideration by any other journal. All authors declare no competing financial interests. The use of generative AI tools in the research is declared in the manuscript per Elsevier policy. AI tools are *not* listed as authors and bear no responsibility for editorial judgements.

Suggested reviewers.

- **Wil M. P. van der Aalst** — TU Eindhoven, foundational PM researcher (w.m.p.v.d.aalst@tue.nl)
- **Marcelo Sergio Pavlovic** — LASIGE, FCUL, PM in software engineering
- **Maria Teresa Baldassarre** — University of Bari, empirical SE and SLRs
- **Eugene Syriani** — University of Montreal, LLM-assisted SLR methodology
- **Tore Dybå** — SINTEF / University of Oslo, evidence-based SE

Reviewers to avoid. None on basis of co-authorship within the past five years. Specific exclusions: thesis advisors and committee members at PUCPR.

Sincerely,

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Paraná (PUCPR)